Total No. of Questions :4 ]

SEAT No. :

**P8607**

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**Oct-22 /TE /Insem - 610**

**T.E. (Electronics & Computer Engineering)**

**MICROCONTROLLER & APPLICATIONS**

**(2019 Pattern) (Semester - I) (310344)**

*Time : 1 Hour]* *[Max. Marks : 30*

***Instructions to the candidates:***

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Figures to the right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Assume suitable data, if necessary.*

***Q1)*** a) Explain various addressing modes of 8051 with suitable examples. **[8]**

b) Explain in detail SFRs related with serial communication in 8051 **[7]**

OR

***Q2)*** a) Explain the use of following pins of 8051 **[8]**

i) ALE

ii) $\overline{EA}$

iii) TXD

iv) RST

b) Explain in detail Interrupt Structure of 8051.State interrupt priority upon reset **[7]**

***Q3)*** a) Draw an interfacing diagram of 4 × 4 Matrix keyboard with 8051. Write an algorithm to detect key pressed. **[8]**

b) Explain in roles of **[7]**

i) Debugger

ii) Emulator &

iii) Assembler

in the development of microcontroller based systems

OR

***Q4)*** a) Draw neat diagram to interface seven segment display (Common Anode) with 8051 & write an embedded 'C' program to display 0–9 number on seven segment display Assume Suitable delay. **[8]**

b) Explain various control signals used in LCD interfacing with 8051. **[7]**